

Ex: $L: \mathbb{R}^2 \rightarrow \mathbb{R}$ is $L(0,1)=2$ enough info?
 $L(0,3)=6$

if yes then $(a,b) = k(0,1) + l(0,3) = (0, k+3l)$
 $a=0$ $b=k+3l$

if the first equation is true then a must be 0
most points in $\mathbb{R}^2$ will have $a \neq 0$

the given info is not enough info to know all outputs

the pattern $ax+by$ or ax^2+bx+c $\in \mathbb{P}_2$
 $\{x^2, x^1, x^0\} \in \mathbb{P}_2$ or $\mathbb{C}^3 \ni (1, 2, 3i)$
||

coefficients come from the field
vectors $1(1,0,0) + 2(0,1,0) + 3i(0,0,1)$

Def: Let V be a vector space over F , let
 $x_1, x_2, \dots, x_n \in V$ and $a_1, a_2, \dots, a_n \in F$. A vector
of the form: $a_1x_1 + a_2x_2 + \dots + a_nx_n$ is called
a linear combination. Superposition!

Warning!: every vector can be expressed as a linear
combination in more than one way.

$$(2,3) = k(1,1) + l(1,-1) = \frac{5}{2}(1,1) - \frac{1}{2}(1,-1)$$

$$\text{but also } (2,3) = 2(1,0) + 3(0,1)$$

Which way of writing a vector as a linear combination
depends on the problem at hand

Ex: write $(7, -1)$ as a linear combination of
 $\{(1, 3), (-1, 2), (1, 1)\}$

$$(7, 1) = \alpha(1, 3) + \beta(-1, 2) + \gamma(1, 1)$$

$$\begin{aligned} 7 &= \alpha - \beta + \gamma \\ 1 &= 3\alpha + 2\beta + \gamma \end{aligned} \quad \text{one equation too few}$$

$$7 = \alpha - \beta + \gamma$$

$$-1 = 3\alpha + 2\beta + \gamma$$

$$1 = -3\alpha - 2\beta - \gamma$$

$$7 = \alpha - \beta + \gamma$$

$$8 = -2\alpha - 3\beta$$

$$\alpha = \frac{-3\beta - 8}{2} = \left[-\frac{3}{2}\beta - 4 \right] = \alpha$$

$$7 = \left(-\frac{3}{2}\beta - 4 \right) - \beta + \gamma$$

$$= -\frac{5}{2}\beta + \gamma - 4$$

$$\left[\gamma = \frac{5}{2}\beta + 11 \right]_{\beta}$$

$$-1 = 3\left(-\frac{3}{2}\beta - 4 \right) + 2\beta + \left(\frac{5}{2}\beta + 11 \right)$$

$$-1 = -\frac{9}{2}\beta - 12 + 2\beta + \frac{5}{2}\beta + 11$$

$-12 = -12$ every value of β satisfies this eqn.

$$(7, -1) = \left(-\frac{3}{2}\beta - 4 \right)(1, 3) + \beta(-1, 2) + \left(\frac{5}{2}\beta + 11 \right)(1, 1)$$

for ex: $\beta = 0$, $(7, -1) = (-4)(1, 3) + (0)(-1, 2) + 11(1, 1) \checkmark$

$$\beta = 2 \quad (7, -1) = (-7)(1, 3) + 2(-1, 2) + 16(1, 1)$$

yes we can express this as a linear combination
of 3 vectors